A Closed-Form Model for the IEEE 802.3az Network and Power Performance

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***Abstract*—We propose an analytical model able to accurately estimate both power consumption and network performance indexes of Energy Efficient Ethernet (EEE) links working at the three available speeds under various traffic load patterns and packet size distributions. The model is sufficiently flexible and accurate to consider different traffic parameters; among others, the packet size distribution, the average burst inter-arrival rate, the burst size distribution, etc. With relatively low complexity, since it addresses stationary queue behavior, the analysis allows obtaining the average energy consumption of the link and, unlike previous works, the mean latency time experienced by incoming packets in closed form, without any upper or lower bound approximations. This aspect makes the model suitable to be adopted in optimization frameworks for network design and control purposes. The numerical results of the model are validated against measurements on a real test bench.**

***Index Terms*—energy management, ethernet networks, queue modeling**

I. Introduction

IN RECENT years, energy consumption has become one of the most important issues in the Information and Communication Technology (ICT) field for both economic and environmental aspects. During the last decade, the energy consumption and the carbon footprint due to ICT have been exponentially increasing, owing to the wide spreading of broadband network access, the wide market penetration of consumer electronics products (e.g., smartphones, tablets, etc.), and the enlargement of datacenters for providing new (cloud) mass e-services.

Today, ICT is accounted to cause 3–5% of greenhouse gas emission, and the energy consumption of ICT infrastructures has become the main cost (between 50% and 70%) in operating expenses of Telecom and service providers [1]–[3]. Moreover, these figures may get even less sustainable in the next years, since several studies indicate that energy cost will rise strongly over the next coming decades.

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However, this situation is not odd, since, except for battery-supplied electronic devices, energy efficiency has never been considered as a design constraint till today. In fact, the energy consumption of the largest part of network hardware (from entire routers and switches to host line cards) is basically flat, and it does not depend on their actual workload [2]. This flat energy consumption is substantially due to the complete absence of mechanisms and solutions for optimizing power absorption in network hardware, which is generally designed with the single objective to provide the maximum performance in any operating conditions (also when idle or under very low utilization).

On the other side, networks are well known to be highly under-utilized with respect to their capacity. For instance, network interfaces at hosts are unused most of the time, with average load ranging from 5% for a typical computer up to 30% for busy servers [4]. In addition, actual traffic is subject to day/night profiles, which makes interesting and suitable to adjust the consumption on the basis of the traffic load [2].

A well-known and effective approach for saving energy consists of introducing advanced power management schemes, which allow consuming energy proportionally to the actual workload of the device. In this respect, the IEEE 802.3az Energy Efficient Ethernet (EEE) standard [4]–[6] is one of the first link-level protocols that exploit power management mechanisms to dynamically reduce the energy consumption according to the traffic load. In more detail, it defines active and sleeping modes for the link, and introduces a technique to coordinate transitions of these modes between the Physical/Medium Access Control (PHY/MAC) chips at the two link terminations. Moreover, it specifies Low Power Mode (LPM) for widely used physical layers of Ethernet over Unshielded Twisted Pair (UTP), namely 100BASE-TX (100Mbps), 1000BASE-T (1Gbps), and 10GBASE-T (10Gbps).

Unfortunately, the adoption of power management techniques, like LPM or Adaptive Rate (AR) [7], often affects network performance in terms of increased transmission delays. For example, the transitions to and from LPM require non-negligible times, which are needed to put network hardware to sleep and to wake it up. Such transition times are often comparable or even much larger than the duration of packet/frame[1] transmissions [6]. This is the case with the IEEE 802.3az standard.

[1]For the purpose of this paper the terms packet and frame are conceptually interchangeable.

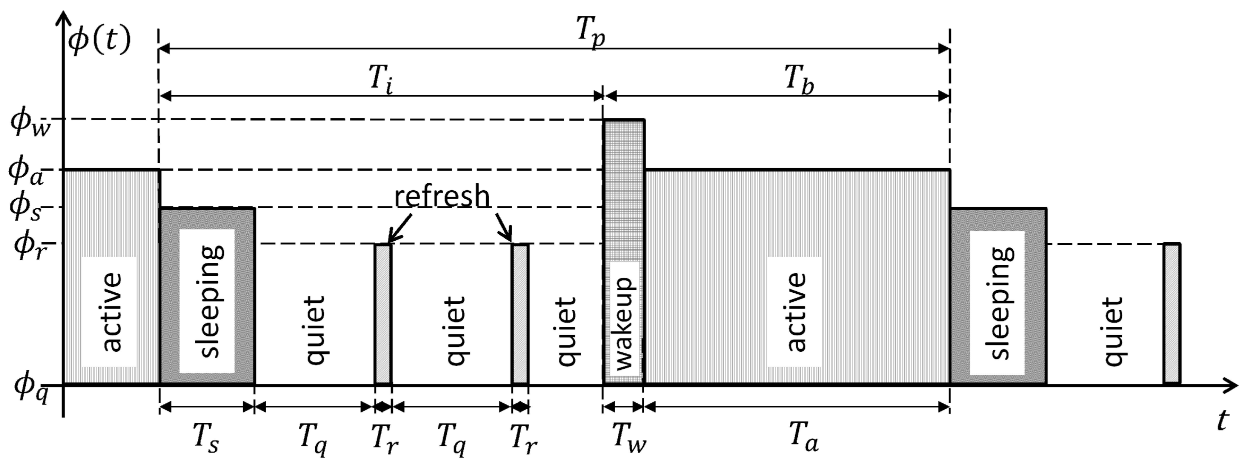

Fig. 1. System cycle and relative power consumption of 100BASE-TX, 1000BASE-T, and 10GBASE-T EEE links.

Starting from these observations, it is clear that, especially in environments requiring tight Quality of Service (QoS) requirements like datacenters, the complex relationships between network performance and energy consumption need to be carefully considered and evaluated.

To this purpose, a number of sophisticated analytical models [8]–[10] and performance evaluation analyses [5], [11] have been recently proposed. However, despite these efforts, a general model able to precisely capture the behavior of the IEEE 802.3az standard when working in the three available versions (i.e., 100BASE-TX, 1000BASE-T, and 10GBASE-T) under variable packet sizes, is still missing. Moreover, most of the proposed analytical models devoted to analyze the energy issues do not consider metrics related with network performance (e.g., packet latencies).

In particular, the authors in [8] provide an accurate analytical model for the burst transmission method that can be applied to further optimize the energy consumption of EEE interfaces. By varying its conguration parameters, like the maximum allowed queue size and the maximum added queuing time, the model helps analyze the expected power saving for different load averages and line card physical characteristics, i.e., nominal link speed, transition times, etc. However, this model is valid only for the unidirectional cases (100BASE-TX and 10GBASE-T links), while it is not applicable for the 1000BASE-T case where the links can enter LPM only when there is no traffic in both directions. Also the work in [9] provides a model to analyze the influence of EEE configuration parameters and interface physical characteristics on the expected performance. Nevertheless, the model is specifically designed for 10GBASE-T links. In [10], the authors provide a model that can be used with 100BASE-TX, 1000BASE-T and 10GBASE-T EEE links, but only refers to the energy consumption, without considering the estimation of network performance indexes.

Starting from these considerations, in this paper we propose an analytical model able to accurately estimate both power consumption and network performance indexes of EEE links working at the three available speeds, under various traffic load patterns and packet size distributions.

The model is sufficiently flexible and accurate to consider different traffic parameters; among others, the frame size distribution, the average burst arrival rate, the burst size distribution, etc. With relatively low complexity, since it addresses stationary queue behavior, the analysis conducted here allows obtaining the average energy consumption of the link and the mean latency time experienced by incoming frames in closed form, without any upper or lower bound approximations. This aspect makes the model suitable to be adopted in optimization frameworks for network design and control purposes. An example of a practical use of the proposed model can be during the planning and/or design phases of a datacenter, where it is necessary to dimension the network interfaces taking into consideration the trade-off between performance and power consumption. In this way, the network architect can size the optimum number of network interfaces per switch, in order to guarantee given delays defined with specific Service Level Agreements (SLAs) and, at the same time, minimize the energy consumed and reduce the costs. It is worth noting that the computational complexity of our model is comparable (if not lower) to that of other, less detailed, models in the literature, owing to its closed form solution. As such, it is reasonable to think of its possible adoption even in on-line feedback control of the trade-off between power consumption and performance based on parametric optimization (e.g., by choosing the duration of the coalescing interval and the number of packets/burst in burst transmission methods), which is the subject of future investigation.

TABLE I
MINIMUM/MAXIMUM SLEEP, QUIET, REFRESH, AND WAKE-UP TIMES FOR DIFFERENT SPEEDS

Protocol	T_s	T_q	T_r	T_w
100BASE-TX	90 / 100 μs	18 / 22 ms	198 μs	27 / 30 μs
1000BASE-T	180 / 200 μs	20 / 22 ms	198 μs	16 μs
10GBASE-T	2.88 μs	20 / 22 ms	198 μs	4.16 μs

The rest of this paper is organized as follows. Subsection I-A introduces the EEE and the behavior of the links. The proposed model is described in Section II. Numerical results and model validation by comparison with measurements on a real test bench are shown in Section III. Finally, the conclusions are drawn in Section IV.

TABLE II
NOTATION DEFINITION

M	dimension of the finite set of discrete service rates.
T_s	sleep time required to enter the LPM.
T_q	LPM or quiet time.
T_r	refresh time needed to maintain the alignment between TX and RX.
T_w	wake-up time required to exit the LPM.
ϕ_a	power consumption during frame transmission.
ϕ_s	power consumption during sleep time.
ϕ_q	power consumption during LPM (quiet time).
ϕ_r	power consumption during refresh time.
ϕ_w	power consumption during wake-up time.
λ	average rate of batch arrival.
μ_i	i^{th} service rate.
m_i	probability that an incoming packet is served with μ_i.
$\mu_{(1)}$, $\mu_{(2)}$	first and second moment of the service rate distribution.
β_j	probability that an incoming burst contains j packets.
$X(z)$	Probability Generating Function (PGF) of the batch size.
j_{max}	maximum number of frames in a burst.
$\beta_{(1)}$, $\beta_{(2)}$	first and second moment of the burst size distribution (in terms of number of frames).
v_t	probability to have a setup period of duration t.
$v_{(1)}$	first moment of the vacation time distribution.
ρ	traffic utilization of the server, which can be expressed as $\rho = \lambda \cdot \beta_{(1)}/\mu_{(1)}$ in the case of infinite buffer.
$B(s)$	Laplace Stieltjes transform of the service time process.
$V(s)$	Laplace Stieltjes transform of the setup time process.
T_i	average duration of server idle periods, including T_s.
T_b	average duration of server busy periods.
T_p	average duration of the idle/busy renewal process.

A. Energy Efficient Ethernet

The approach of EEE is to limit transmission when there is no data, by only maintaining the alignment between Transmitter (TX) and Receiver (RX) with short periodic refresh pulses [6]. The standard defines mechanisms to stop transmission when there is no data to send and to resume quickly when new frames arrive. In order to save energy, the concept of Low Power Mode (LPM) was introduced, which is used instead of continuous IDLE signals during the periods of absence of traffic.

Fig. 1 shows a state transition example of a given link that follows the EEE standard. In summary, T_s is the sleep time defined as the time required to enter LPM, while T_w refers to the wake-up time required to exit LPM. The LPM period is defined as T_q (quiet time). Finally, in order to align TX and RX, EEE considers the short period T_r (refresh time). This periodic state during LPM is used to keep the clock circuitry and other parameters in synchronization before going back to LPM. Refresh permits a faster wake-up time.

Frame transmissions occur during the active periods (T_a). When the TX finishes the transmission of all frames in the queue, it enters the sleep period. When no frames arrive for T_s seconds, it enters the quiet period. A frame arrival during this last period causes the TX to enter the wake-up transition time for T_w seconds and, then, enter again the active period to start the transmission.

It is worth remembering that there are some differences between 100BASE-TX, 1000BASE-T and 10GBASE-T. In EEE 100BASE-TX and 1000BASE-T links, a frame arrival during the sleep time causes a transition to the active state, without passing through the wake-up time. So, in this case, the sleeping action is interruptible, to avoid delays for large values of T_s. On the contrary, the immediate interruption of the sleep time is not supported by EEE 10GBASE-T links. For this reason, the standard value of T_s is less than 3 μs [10]. Moreover, the non-interruption of the sleep time requires buffering the incoming frames[2]. Another important difference is that this mechanism is defined for unidirectional links for 100BASE-TX and 10GBASE-T, while 1000BASE-T links can enter LPM only when there is no traffic for a period T_s in both directions.

Fig. 1 also indicates a qualitative relationship between the energy consumption of the various periods (ϕ_a, ϕ_w, ϕ_s, ϕ_r and ϕ_q for T_a, T_w, T_s, T_r and T_q, respectively). Instead, Table I shows the reference values of T_s, T_q, T_r, and T_w for different speeds as defined in the IEEE 802.3az standard [6].

Unfortunately, as can be noted from Table I, the sleep and wake-up times are considerably higher with respect to the frame transmission time T_f, especially when the frame is small (e.g. 60 bytes, excluding the 4 bytes of Frame Check Sequence) [11], [12] – even by 1 or 2 orders of magnitude. Indeed, in this last case, the saving is low and most of the time and energy are spent during the transition times. In particular, the energy overhead is high for small frames and high speeds.

[2]Later on, in the model description, we neglect the non-interruptible sleeping time for the 10GBASE-T, since the T_s value is very low.

II. The Model

We model the IEEE 802.3az as a couple of independent M^x/D$_M$/1/SET queuing systems, one for each link direction. For simplicity purposes, we label the two directions (and then the two queuing systems) as "Reception" (RX) and "Transmission" (TX). In each queuing system, the traffic arrives in batches of randomly distributed frames, whose service times can assume a discrete set of M values, depending on their possible sizes [13], [14]. The distributions and the average values of such parameters can obviously differ between the RX and TX directions.

The two queuing systems also include stochastic setup times (SET), in order to represent the case in which an incoming frame experiences the wake-up delay due to link LPM-active transitions, or the case where it is immediately served, since it finds the link still active.

Starting from this model, we derive the average latency times experienced by traffic in both the RX and TX directions, and the energy consumption when LPM acts independently on each direction (as in the 100BASE-TX and in the 10GBASE-T IEEE 802.3az versions), or when LPM depends on the state of both directions (the 1000BASE-T case). The remainder of this section is organized as follows. Subsection II-A introduces the core probability generating function and derives some first performance indexes (like the average number of frames in the buffer and the mean latency times) for a single M^x/D$_M$/1/SET queuing system. Since this part of the model is common to both link directions, we decided to drop the "RX" and "TX" indications in the notation. The model of the energy consumption of the link is reported in Subsection II-B, which firstly focuses on deriving some base parameters like idle and busy server times, the probabilities of having LPM transitions, etc. A summary of the notation used in this paper is in Table II.

A. The Single Direction Model

Frames arrive to the link in batches with exponentially distributed inter-arrival times with average rate λ. We denote the probability that an incoming batch contains j frames as β_j. In addition, we assume that the batch size cannot exceed j_{max} frames. So, the Probability Generating Function (PGF) of batch sizes can be expressed as:

$$X(z) = \sum_{j=1}^{j_{max}} \beta_j z^j \tag{1}$$

with the constraint $\sum_{j=1}^{j_{max}} \beta_j = 1$. Therefore, our incoming traffic is modeled as a special case of Batch Markov Arrival Process (BMAP), which has been used as a reasonable representation of real Internet traffic [15], [16]. Depending on the specific characterization of the probability distribution $\beta_j,\ j = 1, \ldots, j_{max}$, Long Range Dependent (LRD) traffic properties can be taken into account.

Regarding the service process, we assume M possible frame sizes (in bytes). To this purpose, we consider a discrete set of frame service rates $\mu_1, \ldots, \mu_M$, with given associated probabilities $m_1, \ldots, m_M$. We can express the Laplace Stieltjes transform of service times as:

$$B(s) = \int_0^\infty \sum_{i=1}^{M} m_i \cdot \delta\left(t - \frac{1}{\mu_i}\right) e^{-st}\,\mathrm{d}t = \sum_{i=1}^{M} m_i \cdot e^{-\frac{s}{\mu_i}} \tag{2}$$

We derive the PGF of the M^x/D$_M$/1 queuing system from the one of the more general M^x/G/1 system [17], [18]:

$$P\big(z; \mathrm{M}^x/\mathrm{G}/1\big) = (1-\rho)\,\frac{(1-z)\,B\big(\lambda - \lambda X(z)\big)}{B\big(\lambda - \lambda X(z)\big) - z} \tag{3}$$

By using Eqs. 1 and 2 in 3, we obtain:

$$P\big(z; \mathrm{M}^x/\mathrm{D}_M/1\big) = (1-\rho)\frac{(1-z)\sum_{i=1}^{M} m_i e^{-\frac{\lambda}{\mu_i}\left[1-X(z)\right]}}{\sum_{i=1}^{M} m_i e^{-\frac{\lambda}{\mu_i}\left[1-X(z)\right]} - z} \tag{4}$$

In order to consider the presence of server wake-up times, we exploit the stochastic decomposition results of Doshi [19] for the single unit arrival case and the results in [17] for bulk arrivals. The PGF of the M^x/D$_M$/1 queue with setup times turns out to be:

$$P\big(z; \mathrm{M}^x/\mathrm{D}_M/1/\mathrm{SET}\big) = \xi(z)\,P\big(z; \mathrm{M}^x/\mathrm{D}_M/1\big) \tag{5}$$

where

$$\xi(z) = \frac{1 - zV\big(\lambda - \lambda X(z)\big)}{\Big(\frac{1}{\beta_{(1)}} + \lambda v_{(1)}\Big)\left[1 - X(z)\right]} \tag{6}$$

is the PGF of the number of arrivals during the residual life of the vacation period, defined as an idle period plus a setup period. $\beta_{(1)}$ is the first moment of the number of frames per burst. $V(s)$ and $v_{(1)}$ are the Laplace Stieltjes transform and the first moment of the setup time process, respectively. Regarding the setup times, we have to distinguish two possible cases:

1) The batch is received at an empty buffer while the link is in LPM; then, a server setup vacation with a fixed duration T_w is required to turn on the link (see Table I);
2) The batch is received at an empty buffer while the link is still active; then, no server vacation is required, and the frame service process can start immediately.

We denote with v_{T_w} the probability that the incoming batch, causing the server exiting the idle period and entering the setup vacation period, experiences case (1). This probability corresponds to that of the batch finding the link off, while the server is idle. We can express $V(s)$ as:

$$\begin{aligned} V(s) &= \int_0^\infty e^{-st}\left[(1 - v_{T_w})\,\delta(t) + v_{T_w}\delta(t - T_w)\right]\mathrm{d}t \\ &= 1 - v_{T_w}\left(1 - e^{-T_w s}\right) \end{aligned} \tag{7}$$

and, therefore, $v_{(1)} = v_{T_w} \cdot T_w$ (note that in case (2) the setup time is zero).

Thus, by using Eqs. 4, 5, 6, and 7, we can finally write the PGF of the M^x/D$_M$/1/SET queuing system as indicated in Eq. 8.

At this point, the mean value $\bar{L}$ of the number of frames in the queuing system is obtained as shown in Eq. 9, where $\beta_{(2)}$ and $\mu_{(2)}$ are the second moments of the number of frames in a burst and of the service rate, respectively, and $v_{(2)} = v_{T_w} T_w^2$.

$$P\big(z;\mathrm{M}^x/\mathrm{D}_M/1/\mathrm{SET}\big) = \frac{1 - z\left(v_{T_w} e^{-T_w\lambda\left(1-X(z)\right)} + 1 - v_{T_w}\right)}{\left(\frac{1}{\beta_{(1)}} + \lambda v_{(1)}\right)\left[1 - X(z)\right]} \cdot \frac{(1-\rho)\,(1-z)\sum_{i=1}^{M} m_i e^{-\frac{\lambda}{\mu_i}\left[1-X(z)\right]}}{\sum_{i=1}^{M} m_i e^{-\frac{\lambda}{\mu_i}\left[1-X(z)\right]} - z} \tag{8}$$

$$\bar{L} = \lim_{z\to 1} \frac{dP\big(z;\mathrm{M}^x/\mathrm{D}_M/1/\mathrm{SET}\big)}{dz} = \frac{2\lambda\beta_{(1)}v_{(1)} + \lambda^2\beta_{(1)}^2 v_{(2)} + \lambda v_{(1)}\left(\beta_{(2)} - \beta_{(1)}\right)}{2\left(1 + \lambda\beta_{(1)}v_{(1)}\right)} + \frac{\frac{\lambda^2\beta_{(1)}^2}{\mu_{(2)}} + \frac{\lambda}{\mu_{(1)}}\left(\beta_{(2)} - \beta_{(1)}\right)}{2\,(1-\rho)} \tag{9}$$

Using Little's law, the average packet waiting time $\hat{W}$ of incoming packets is:

$$\hat{W} = \frac{\bar{L}}{\lambda\beta_{(1)}} \tag{10}$$

B. The Energy Consumption Model

As introduced in I-A, the power consumption of the IEEE 802.3az link depends on the status of both the reception (RX) and transmission (TX) directions. Moreover, LPM-active transitions happen independently at RX and TX directions only in the 100BASE-TX and 10GBASE-T versions, while in 1000BASE-T the link can enter LPM only when both directions have been simultaneously idle for T_s seconds or more.

To handle such aspect, we firstly derive the average idle and busy periods of RX and TX servers, and then we use these parameters to express the energy consumption in both cases of 100BASE-TX and 10GBASE-T and in the one of 1000BASE-T.

1) The idle and busy server periods: Under server traffic utilization $\rho = \lambda\beta_{(1)}/\mu_{(1)} < 1$, a G/G/1 queuing system will become empty infinitely often. This obviously remains true also for our M^x/D$_M$/1/SET model. Hence, by using classical principles of renewal theory, we can identify independent and identically distributed (iid) "cycles" of the form:

$$T_p^{(n)} = T_i^{(n)} + T_b^{(n)} \tag{11}$$

where $T_b^{(n)}$ is the nth busy period, (corresponding to the "delay busy period" in [17], which includes the setup time), and $T_i^{(n)}$ is the n^{th} idle period (IDP in [17]). In more detail, both sequences $T_b^{(n)}$ and $T_i^{(n)}$ can be demonstrated to be iid. The average duration of idle and busy periods can be easily shown to be [17]:

$$\begin{aligned} T_i = \mathrm{E}\Big\{T_i^{(n)}\Big\} &= \frac{1}{\lambda} \\ T_b = \mathrm{E}\Big\{T_b^{(n)}\Big\} &= \frac{\beta_{(1)}\left(1/\mu_{(1)} + v_{T_w}T_w\right)}{1-\rho} \end{aligned} \tag{12}$$

We can then obtain T_p as follows:

$$\begin{aligned} T_p &= T_i + T_b \\ &= \frac{1}{\lambda} + \frac{\beta_{(1)}\left(1/\mu_{(1)} + v_{T_w}T_w\right)}{1-\rho} \end{aligned} \tag{13}$$

Thus, the stationary probabilities to be in an idle and busy period, respectively, are:

$$P_i = \frac{T_i}{T_p} = \frac{1-\rho}{1 + \lambda\beta_{(1)}v_{T_w}T_w} \tag{14}$$

$$P_b = \frac{T_b}{T_p} = \frac{\rho + \lambda\beta_{(1)}v_{T_w}T_w}{1 + \lambda\beta_{(1)}v_{T_w}T_w} \tag{15}$$

2) Vacation time distribution: The different LPM transition mechanisms of the IEEE 802.3az versions 100BASE-TX/ 10GBASE-T and of version 1000BASE-T clearly impact on the probabilities of an incoming frame finding the link still active or sleeping, and consequently of experiencing the server wake-up delay or not.

In the case of 100BASE-TX and 10GBASE-T, LPM transitions happen independently at RX and TX servers. So, the probability $v_{T_w}^{(\mathrm{RX})}$ that a frame causing a setup transition to the RX server finds the link sleeping can be written as shown in Eq. 16 below. The $v_{T_w}^{(\mathrm{TX})}$ parameter can be derived in the same way.

Regarding 1000BASE-T, $v_{T_w}^{(\mathrm{RX})}$ corresponds to the probability (conditioned to have the RX server in idle state) that both RX and TX servers have been idle for T_s seconds or more. Thanks to the independence of RX and TX queuing systems, we can re-write $v_{T_w}^{(\mathrm{RX})}$ (and, consequently, $v_{T_w}^{(\mathrm{TX})}$) as indicated in Eq. 17 (Eq. 18).

Then, by substituting Eqs. 18 in 17 and after some algebra, we can write $v_{T_w}^{(\mathrm{RX})}$ and $v_{T_w}^{(\mathrm{TX})}$ for the 1000BASE-T case as in Eqs. 19 and 20.

3) Average energy consumption in 100BASE-TX and 10GBASE-T: The instantaneous energy consumption of an IEEE 802.3az link arises from the status of its two terminations and, in general, it can be expressed as: $\phi(t) =$

$$v_{T_w}^{(\mathrm{RX})} = P\big(\text{RX has been idle for } T_s \text{ seconds or more}\,|\,\text{RX is idle}\big) = \frac{P_i^{(\mathrm{RX})} e^{-\lambda^{(\mathrm{RX})}T_s}}{P_i^{(\mathrm{RX})}} = e^{-\lambda^{(\mathrm{RX})}T_s} \tag{16}$$

$$\begin{aligned} v_{T_w}^{(\mathrm{RX})} &= P\big(\text{RX and TX have been idle for } T_s \text{ or more} \,|\, \text{RX idle}\big) \\ &= P\big(\text{RX has been idle for } T_s \text{ or more} \,|\, \text{RX is idle}\big) \cdot P(\text{TX has been idle for } T_s \text{ or more}) = \\ &= P_i^{(\mathrm{TX})} e^{-\left[\lambda^{(\mathrm{RX})}+\lambda^{(\mathrm{TX})}\right]T_s} = \frac{\left(1-\rho^{(\mathrm{TX})}\right) e^{-\left[\lambda^{(\mathrm{RX})}+\lambda^{(\mathrm{TX})}\right]T_s}}{1+\lambda^{(\mathrm{TX})}\beta_{(1)}^{(\mathrm{TX})} v_{T_w}^{(\mathrm{TX})} T_w} \end{aligned} \tag{17}$$

$$v_{T_w}^{(\mathrm{TX})} = \frac{\left(1-\rho^{(\mathrm{RX})}\right) e^{-\left[\lambda^{(\mathrm{RX})}+\lambda^{(\mathrm{TX})}\right]T_s}}{1+\lambda^{(\mathrm{RX})}\beta_{(1)}^{(\mathrm{RX})} v_{T_w}^{(\mathrm{RX})} T_w} \tag{18}$$

$$\begin{aligned} v_{T_w}^{(\mathrm{RX})} &= \frac{e^{-\left[\lambda^{(\mathrm{RX})}+\lambda^{(\mathrm{TX})}\right]T_s}}{2\beta_{(1)}^{(\mathrm{RX})}\lambda^{(\mathrm{RX})}T_w} \cdot \Bigg\{ \left[\beta_{(1)}^{(\mathrm{RX})}\lambda^{(\mathrm{RX})}T_w\left(1-\rho^{(\mathrm{TX})}\right) - \beta_{(1)}^{(\mathrm{TX})}\lambda^{(\mathrm{TX})}T_w\left(1-\rho^{(\mathrm{RX})}\right) - e^{-\left[\lambda^{(\mathrm{RX})}+\lambda^{(\mathrm{TX})}\right]T_s}\right] + \\ &+ \sqrt{4\beta_{(1)}^{(\mathrm{RX})}\lambda^{(\mathrm{RX})}T_w\left(1-\rho^{(\mathrm{TX})}\right)e^{-\left[\lambda^{(\mathrm{RX})}+\lambda^{(\mathrm{TX})}\right]T_s} + \left[\beta_{(1)}^{(\mathrm{TX})}\lambda^{(\mathrm{TX})}T_w\left(1-\rho^{(\mathrm{RX})}\right) - \beta_{(1)}^{(\mathrm{TX})}\lambda^{(\mathrm{TX})}T_w\left(1-\rho^{(\mathrm{RX})}\right) + e^{-\left[\lambda^{(\mathrm{RX})}+\lambda^{(\mathrm{TX})}\right]T_s}\right]^2} \Bigg\} \end{aligned} \tag{19}$$

$$\begin{aligned} v_{T_w}^{(\mathrm{TX})} &= \frac{e^{-\left[\lambda^{(\mathrm{RX})}+\lambda^{(\mathrm{TX})}\right]T_s}}{2\beta_{(1)}^{(\mathrm{TX})}\lambda^{(\mathrm{TX})}T_w} \cdot \Bigg\{ \left[\beta_{(1)}^{(\mathrm{TX})}\lambda^{(\mathrm{TX})}T_w\left(1-\rho^{(\mathrm{RX})}\right) - \beta_{(1)}^{(\mathrm{RX})}\lambda^{(\mathrm{RX})}T_w\left(1-\rho^{(\mathrm{TX})}\right) - e^{-\left[\lambda^{(\mathrm{RX})}+\lambda^{(\mathrm{TX})}\right]T_s}\right] + \\ &+ \sqrt{4\beta_{(1)}^{(\mathrm{TX})}\lambda^{(\mathrm{TX})}T_w\left(1-\rho^{(\mathrm{RX})}\right)e^{-\left[\lambda^{(\mathrm{RX})}+\lambda^{(\mathrm{TX})}\right]T_s} + \left[\beta_{(1)}^{(\mathrm{TX})}\lambda^{(\mathrm{TX})}T_w\left(1-\rho^{(\mathrm{RX})}\right) - \beta_{(1)}^{(\mathrm{TX})}\lambda^{(\mathrm{TX})}T_w\left(1-\rho^{(\mathrm{RX})}\right) + e^{-\left[\lambda^{(\mathrm{RX})}+\lambda^{(\mathrm{TX})}\right]T_s}\right]^2} \Bigg\} \end{aligned} \tag{20}$$

$\phi^{(\mathrm{RX})}(t) + \phi^{(\mathrm{TX})}(t)$, where:

$$\phi^{(\mathrm{RX})}(t) = \phi^{(\mathrm{TX})}(t) = \begin{cases} \phi_a & \text{during active time} \\ \phi_s & \text{during sleep time} \\ \phi_q & \text{during LPM/quiet time} \\ \phi_r & \text{during refresh time} \\ \phi_w & \text{during wake-up time} \end{cases} \tag{21}$$

For the sake of simplicity, let us focus only on the RX side, and fix $\phi_s = \phi_a$. Recalling Fig. 1 and Eq. 21, we can express the average power consumption $\tilde{\phi}^{(\mathrm{RX})}$

$$\tilde{\phi}^{(\mathrm{RX})} = \frac{\phi_a T_a^{(\mathrm{RX})} + \tilde{E}_i^{(\mathrm{RX})} + \phi_w v_{T_w}^{(\mathrm{RX})} T_w}{T_p^{(\mathrm{RX})}} \tag{22}$$

where:

- $T_a^{(\mathrm{RX})}$ is the average time in which the RX server is busy, excluded the time spent during setup vacations; then, $T_a^{(\mathrm{RX})} = T_b^{(\mathrm{RX})} v_{T_w}^{(\mathrm{RX})}$.
- $\tilde{E}_i^{(\mathrm{RX})}$ is the average energy consumed during RX server idleness.

In more detail, $\tilde{E}_i^{(\mathrm{RX})}$ includes sleeping periods and related link refresh operations, as well as the periods where the link is maintained active after the buffer becomes empty. It can be decomposed as:

$$\tilde{E}_i^{(\mathrm{RX})} = \tilde{E}_s^{(\mathrm{RX})} + \tilde{E}_q^{(\mathrm{RX})} + \tilde{E}_r^{(\mathrm{RX})} \tag{23}$$

where $\tilde{E}_s^{(\mathrm{RX})}$, $\tilde{E}_q^{(\mathrm{RX})}$, $\tilde{E}_r^{(\mathrm{RX})}$ represent the average amount of energy spent before entering LPM, during LPM and during refresh periods, respectively. Recalling that batch inter-arrival times follow an exponential distribution, and given that sleeping periods are interruptible if a new batch arrives at the RX server before T_s, we can simply obtain $\tilde{E}_s^{(\mathrm{RX})}$ as the sum of two contributions representing an interrupted and a non-interrupted sleeping period, respectively. Taking this in mind, $\tilde{E}_s^{(\mathrm{RX})}$ turns out to be:

$$\begin{aligned} \tilde{E}_s^{(\mathrm{RX})} &= \int_0^{T_s} \phi_a t \lambda^{(\mathrm{RX})} e^{-\lambda^{(RX)}t}\,\mathrm{d}t + \\ &\quad \phi_a T_s \int_{T_s}^{\infty} \lambda^{(\mathrm{RX})} e^{-\lambda^{(\mathrm{RX})}} t\,\mathrm{d}t \\ &= \frac{\phi_a}{\lambda^{(\mathrm{RX})}}\left[1 - e^{-\lambda^{(\mathrm{RX})}T_s}\left(1+\lambda^{(\mathrm{RX})}T_s\right)\right] + \\ &\quad \phi_a T_s e^{-\lambda^{(\mathrm{RX})}T_s} \end{aligned} \tag{24}$$

Regarding the average energy consumed during refresh operations, it simply depends on the mean number N_r of times the link is refreshed in a renewal period:

$$\tilde{E}_r^{(\mathrm{RX})} = \phi_r T_r N_r \tag{25}$$

The mean number of link refreshes can be derived as the average number of $T_q + T_r$ cycles in the residual server idle time following a sleeping time, as shown in Eq. 26.

Finally, we can write the average energy spent in LPM as ϕ_q multiplied by the idle period minus the sleeping and the

$$
\begin{aligned}
N_r = \mathrm{E}\Big\{N_r^{(n)}\Big\} &= \mathrm{E}\left\{\left\lfloor \frac{T_i^{(n)} - T_s^{(n)}}{T_q + T_r} \right\rfloor\right\} = \int_{T_s}^{\infty} \left\lfloor \frac{t - T_s}{T_q + T_r} \right\rfloor \lambda^{(\mathrm{RX})} e^{-\lambda^{(\mathrm{RX})} t}\, \mathrm{d}t \\
&= \sum_{f=1}^{\infty} f \int_{f\left(T_q+T_r\right)+T_s}^{(f+1)\left(T_q+T_r\right)+T_s} \lambda^{(\mathrm{RX})} e^{-\lambda^{(\mathrm{RX})} t}\, \mathrm{d}t = \frac{e^{-\lambda^{(\mathrm{RX})} T_s}}{\left[e^{-\lambda^{(\mathrm{RX})}\left(T_q+T_r\right)} - 1\right]}
\end{aligned} \tag{26}
$$

refresh times:

$$
\begin{aligned}
\tilde{E}_q^{(\mathrm{RX})} &= \phi_q \left[\int_{T_s}^{\infty} (t - T_s)\, \lambda^{(\mathrm{RX})} e^{-\lambda^{(\mathrm{RX})} t} dt - T_r N_r\right] \\
&= \phi_q e^{-\lambda^{(\mathrm{RX})} T_s} \cdot \\
&\quad \left[\frac{1}{\lambda^{(\mathrm{RX})}} - \frac{T_r}{e^{-\lambda^{(\mathrm{RX})}\left(T_q+T_r\right)} - 1}\right]
\end{aligned} \tag{27}
$$

4) Average energy consumption in 1000BASE-T: In the case of a 1000BASE-T link, we maintain Eqs. 21 and 22 and related definitions from the previous subsection. Here, the main difference comes from the contribution $\tilde{E}_i^{(\mathrm{RX})}$, since the link's LPMs depend also on the status of the TX server. In more detail, the link enters LPM only when both servers have been idle for T_s or more, and exits when a batch is received by one of the two directions. The aggregated batch arrivals still follow a Poisson process with rate $\lambda^{(\mathrm{RX})} + \lambda^{(\mathrm{TX})}$. Recalling the definition of $v_{T_w}^{(\mathrm{RX})}$ from Eq. 17 and the above considerations, we can easily deduce that:

$$
\tilde{E}_i^{(\mathrm{RX})} = v_{T_w}^{(\mathrm{RX})} \tilde{E}_{LPM}^{(\mathrm{RX})} + \left(1 - v_{T_w}^{(\mathrm{RX})}\right) \frac{\phi_a}{\lambda^{(\mathrm{RX})} + \lambda^{(\mathrm{TX})}} \tag{28}
$$

where $\tilde{E}_{LPM}^{(\mathrm{RX})}$ is the average energy consumed when the link is in LPM. In other words, the link enters LPM with the same probability that the next incoming frame will experience a non-zero setup vacation. In a very similar way with respect to Eqs. 26 and 27, and thanks to the memoryless property of Poisson arrivals, we can write that:

$$
N_r = \frac{e^{-\left(\lambda^{(\mathrm{RX})} + \lambda^{(\mathrm{TX})}\right) T_s}}{\left[e^{\left(\lambda^{(\mathrm{RX})} + \lambda^{(\mathrm{TX})}\right)\left(T_q+T_r\right)} - 1\right]} \tag{29}
$$

$$
\begin{aligned}
\tilde{E}_q^{(\mathrm{RX})} &= \left[\frac{\phi_q e^{-\left(\lambda^{(\mathrm{RX})} + \lambda^{(\mathrm{TX})}\right) T_s}}{\lambda^{(\mathrm{TX})} + \lambda^{(\mathrm{RX})}} - \right. \\
&\quad \left. \frac{T_r \phi_q e^{-\left(\lambda^{(\mathrm{RX})} + \lambda^{(\mathrm{TX})}\right) T_s}}{e^{\left(\lambda^{(\mathrm{RX})} + \lambda^{(\mathrm{TX})}\right)\left(T_q+T_r\right)} - 1}\right]
\end{aligned} \tag{30}
$$

Finally, we can obtain $\tilde{E}_{LPM}^{(\mathrm{RX})}$ as:

$$
\tilde{E}_{LPM}^{(\mathrm{RX})} = \tilde{E}_q^{(\mathrm{RX})} + \phi_r T_r N_r \tag{31}
$$

III. Model validation

We validated the proposed model against real-world 15-minute-long traffic traces of 1Gbps links obtained from the MAWI (Measurement and Analysis on the WIDE Internet) Archive [20] for both the unidirectional and bidirectional cases. Unfortunately, the traffic volumes of these traces present values too high to enter the LPM mode of the IEEE 802.3az switches used in our test bench. For this reason, in order to guarantee a significant validation of the model, we decrease the traffic volumes in the original trace by a scaling factor of 4.

The rest of this section describes the test bench used in our validation and the configuration for the two different cases: unidirectional (100BASE-TX and 10GBASE-T links), and bidirectional (1000BASE-T). In addition, in the next section, we present some numerical results for the two cases.

A. Test bench

The test bench used in our model validation is shown in Fig. 2. It involves three main elements: two connected switches EEE 1000BASE-T (Dlink DGS-1005A), an Ixia N2X Router Tester (RT) to generate and receive the traffic trace, and the DC power measurement instrument.

The unidirectional case is realized by emulating the LPM behavior of 100BASE-TX and 10GBASE-T switches (which is driven by each link direction independently of the other) on the 1000BASE-T switch, which is forced to send traffic only in one direction. Indeed, the packet generator transmits the recorded traffic traces using only link A in Fig. 2.

Instead, for the bidirectional case, the standard behavior of the EEE 1000BASE-T switch has been exploited. In this case, the packet generator sends packets in both directions (links A and B in Fig. 2).

The power consumption is measured by collecting samples of the supply voltage and of the current absorption of the switches (Switch 1 in Fig. 2). In detail, we estimated the current absorption by means of very small (10 mΩ) precise Current Sensing Resistors (CSR). The voltage drop across the CSR is proportional to the incoming current. The CSR voltage drops and the supply voltage are finally measured by an external Data AcQuisition (DAQ) device. In our tests, we used an Agilent U2353A multifunction Data AcQuisition (DAQ), which can simultaneously sample 4 channels at 250 kHz with 16-bit resolution.

In the next two subsections, we describe the traffic traces used in our validation and the model parameter settings for the unidirectional and bidirectional cases.

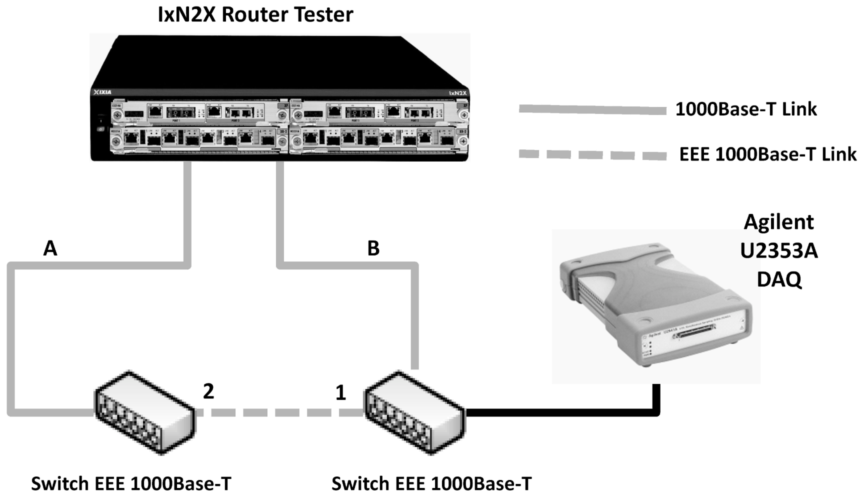

Fig. 2. Reference Scenario for the model validation.

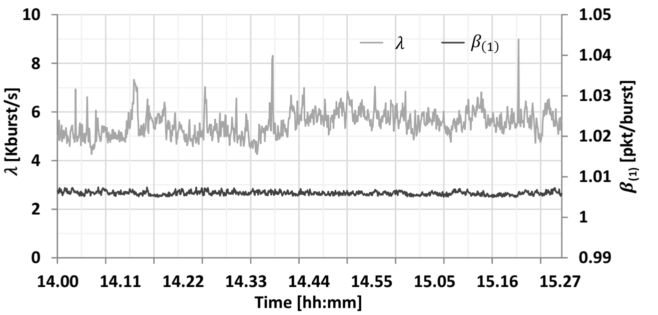

Fig. 3. Average values of λ and $\beta_{(1)}$ measured from the traffic trace in [20].

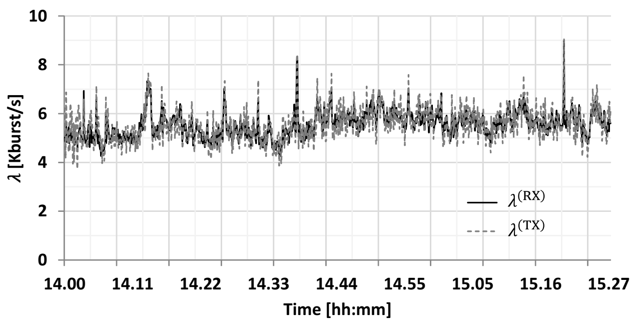

Fig. 4. Average values of λ measured in the traffic trace from [20] and used for the bidirectional case.

B. Traffic Traces

Fig. 3 shows the traffic offered load characterization in burst arrival rates (λ) and burst sizes ($\beta_{(1)}$), which were calculated by starting from the packet level trace, and averaging the data over different time slices. Each time slice has a fixed duration equal to 5 s. The trend of the parameter λ is normally comprised between 4 and 7 kburst per second. When there are traffic spikes λ also reaches 9 kburst per second. Considering that $\beta_{(1)}$ is slightly greater than 1, the offered load in terms of packets per second has a similar trend to that of λ.

For the bidirectional case, Figs. 4 and 5 show the traffic offered load representation in terms of burst arrival rates (λ) and burst sizes ($\beta_{(1)}$), respectively. For the validation in the bidirectional case, we used two different traffic traces: one for the RX and another for the TX. These two traffic traces have similar values of burst arrival rates. Instead, the differences for burst trace sizes are slightly more significant. Nevertheless, as shown in Fig. 5, in both cases the average burst size is not much higher than 1.

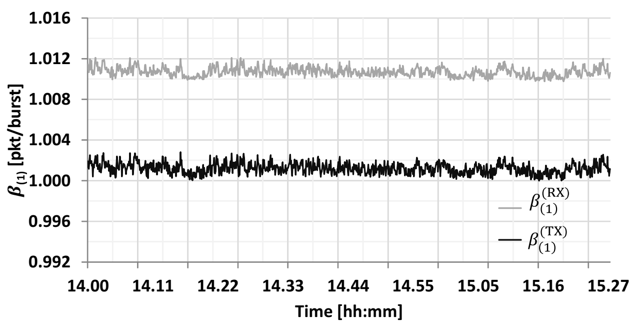

Fig. 5. Average values of $\beta_{(1)}$ measured in the traffic trace from [20] and used for the bidirectional case.

C. Configuration Parameters

Besides the average values of λ and $\beta_{(1)}$, we obtained the estimation of frame size probabilities and the maximum number of frames in a burst j_{max}. Table III shows the most significant values of the frame size distribution. For a complete description of this distribution, see [20].

In this way, we can model the discrete set of service rates of our EEE link by considering the transmission speed and the frame size distribution. Thus, it is reasonable to express the service time $\frac{1}{\mu_j}$ corresponding to each frame

$$\frac{1}{\mu_j} = \frac{C}{f_j \cdot 8} + k \tag{32}$$

where C represents the link capacity (in bps), f_j is the frame

TABLE III
FRAME SIZE DISTRIBUTION MEASURED FROM THE TRAFFIC TRACE IN [20].

Frame size[Byte]	Distribution(%)
60	17.90%
62	1.14%
66	11.09%
74	2.16%
86	1.35%
1294	1.72%
1434	9.82%
1440	2.27%
1494	1.29%
1514	26.86%
others (<1%)	24.41%

TABLE IV
CONFIGURATION PARAMETERS SETTING FOR MODEL VALIDATION.

Parameter	Value
T_w	16μs
T_s	200μs
T_r	198μs
T_q	22ms
ϕ_a	1.2W
ϕ_w	1.2W
ϕ_r	1.2W
ϕ_q	0.85W

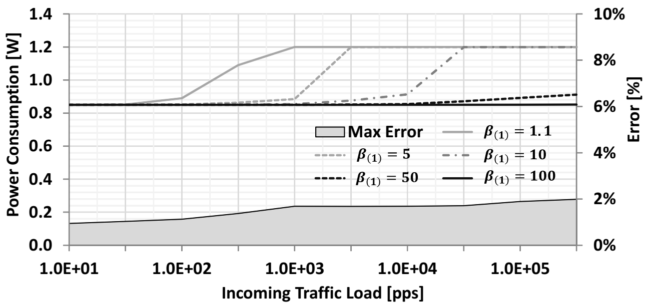

Fig. 6. Average Power consumption measured with the test bench and relative error with respect to the one estimated by the model, by varying the incoming load and the average burst size (in terms of number of packets) $\beta_{(1)}$ in the unidirectional case.

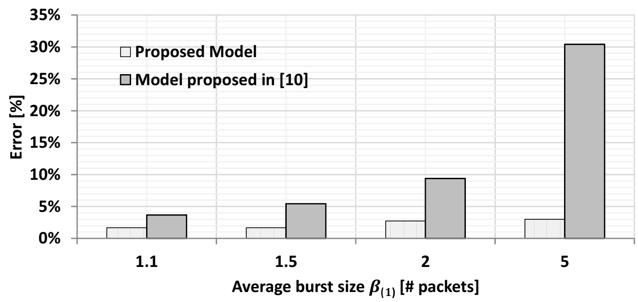

Fig. 7. Maximum estimation error (in %) of the average power consumption for our model and the one proposed in [10] with respect to the values measured with the test bench for different values of the average burst size (in terms of number of packets) $\beta_{(1)}$ in the unidirectional case.

size (in bytes) and k is an additive constant. Based on our measurements, k is equal to 2 μs and it is independent of the link capacity and the frame size. We take the number of frame length values $M = 1514$. The validation tests were performed by estimating and/or measuring the energy consumption and the network performance over successive 1-second time-slices, with both the proposed model and the experiment. For the model, we used the above-mentioned parameters estimated from the traffic trace, whereas, regarding the experimental evaluation, we directly reproduced the packet-level traces. The other configuration parameters are fixed with the values shown in Table IV. Each of the parameters shown in Table III was measured on the test bench of Fig. 2. Obviously, as previously described, the parameters used referred to the EEE 1000BASE-T switch.

D. Numerical Results

Before validating the proposed model with the traffic traces described in the previous subsections, we have tested it by using the test bench described in Subsection III-A and varying the incoming traffic load λ from 10 to 150 kpps with different values of $\beta_{(1)}$ $(1.1, 5, 10, 50$ and $100)$. For the values of the other parameters (service rate, power consumption, etc.) we used the ones described in the previous subsection.

Fig. 6 reports the power consumption measured with the experimental test bench and the maximum error with respect to the values estimated by the model by varying the incoming burst rate λ for the unidirectional case. The results in Fig. 6 show the good accuracy level provided by the model in the unidirectional case. Indeed, the maximum estimation error is always lower than 2%.

Instead, Figs. 7 and 8 show the comparison of the maximum estimation error (in %) of the average power consumption between our model and the one proposed in [10] for different values of $\beta_{(1)}$ and by varying the incoming traffic load between 10 and 150 kpps. Our model has an error always less than 5%, confirming the good accuracy. In more detail, unlike the model proposed in [10], the growth of the estimation error by increasing $\beta_{(1)}$ (Fig. 7) or the incoming traffic load (Fig. 8) is much less significant.

Fig. 9 shows the average values of the packet latency times measured on the test bench and the maximum error with respect to the values estimated by the model, by varying the incoming burst rate λ for the unidirectional case. Also for the packet latency times, the proposed model presents good accuracy level. In this case, the maximum estimation error is always lower than 0.5%.

Finally, in order to complete the validation with respect to varying the incoming traffic load, Fig. 10 shows the trend of the average packet latency times measured with the experimental test bench when EEE is not enabled. We modeled this case by fixing T_s and T_w equal to 0. As for the EEE case, the maximum estimation error is always lower than 1%. After validating the model by varying λ and with different values of $\beta_{(1)}$, in the rest of this subsection we describe the results obtained with the above-mentioned traffic traces.

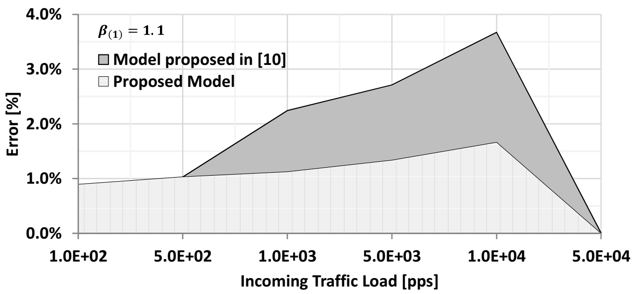

Fig. 8. Estimation error (in %) of the average power consumption for our model and the one proposed in [10] with respect to the values measured with the test bench by varying the incoming traffic load and with the average burst size (in terms of number of packets) $\beta_{(1)} = 1.1$ in the unidirectional case.

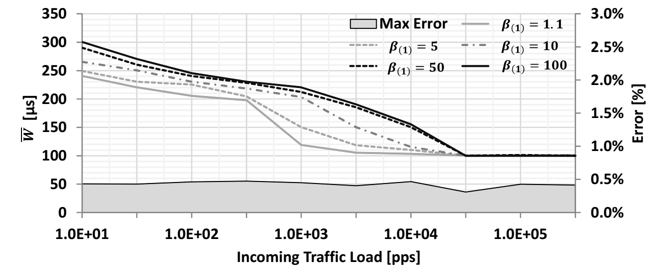

Fig. 9. Average packet waiting time $\hat{W}$ measured with the test bench and relative error with respect to the one estimated by the model, by varying the incoming load and the average burst size (in terms of number of packets) $\beta_{(1)}$ in the unidirectional case.

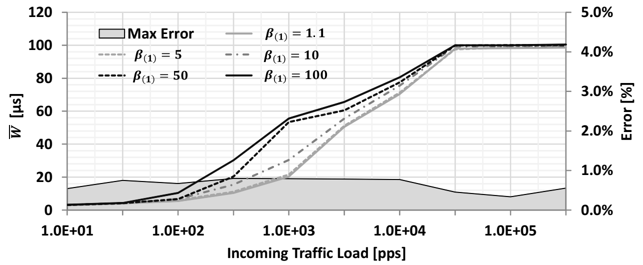

Fig. 10. Average packet waiting time $\hat{W}$ measured with the test bench and relative error with respect to the one estimated by the model, by varying the incoming load and the average burst size (in terms of number of packets) $\beta_{(1)}$ in the unidirectional case without EEE enabled.

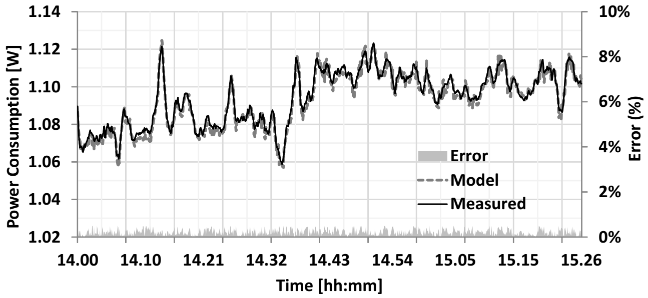

Fig. 11. Average power consumption estimated by the model and measured with the test bench for the unidirectional case.

Fig. 11 reports the power consumption values estimated by the model, the values measured with the experimental test bench, and the maximum estimation error in each time window for the unidirectional case. The model estimations were obtained with Eqs. 22 and 28.

The results in Fig. 11 outline the good accuracy level provided by the model in the unidirectional case. Indeed, the maximum estimation error is always lower than 0.5%.

Fig. 12 shows the average values of the packet latency times estimated by the model (Eq. 10) and measured on the test bench for the unidirectional case. Also for the packet latency times, the proposed model presents good accuracy level. Again, the maximum estimation error is always lower than 0.5%.

The results for the bidirectional case are shown in Figs. 13 and 14. As in the unidirectional case, Fig. 13 reports the power consumption estimated by the model, the values measured with the experimental test bench, and the maximum estimation error in each time window. Fig. 14 shows the average packet latency times with the relative error between the model and the values measured on the test bench. Also in this case, the model provides good accuracy level. Indeed, the maximum estimation error is always lower than 1% for the power consumption (Fig. 13) and lower than 0.5% for the packet latency times (Fig. 14).

IV. Conclusions

We introduced an analytical model able to accurately estimate both power consumption and network performance indexes of EEE links working at the three available speeds under various traffic load patterns and packet size distributions. The complexity of the model has been kept manageable, by considering stationary queue probabilities, rather than the much more complex dynamic evolution in terms of Markov chains. The analysis we have proposed allows obtaining the average energy consumption of the link and the mean latency time experienced by incoming frames in closed form, without any upper or lower bound approximations.

The numerical results of the model were validated against measurements on a real test bench realized with professional network testing and energy consumption measurement instrumentation, under real traffic traces. The results obtained clearly indicate the accuracy of the model (maximum error lower than 1% for the energy consumption and lower than 0.5% for frame latency times).

The availability of an analytical representation of the energy and performance indicators allows the design of control frameworks based on parametric optimization, in order to determine the best tradeoff between power consumption and QoS, according to clearly defined criteria. The determination of such control strategies and their performance evaluation will be the subject of future work.

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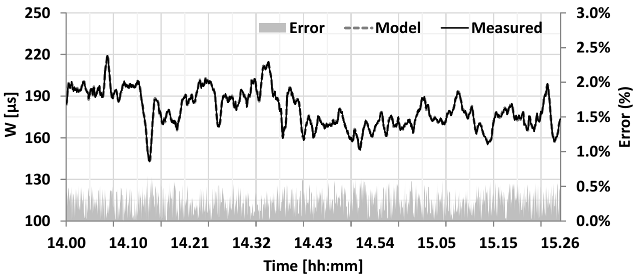

Fig. 12. Average packet waiting time $\hat{W}$ estimated by the model and measured with the test bench for the unidirectional case.

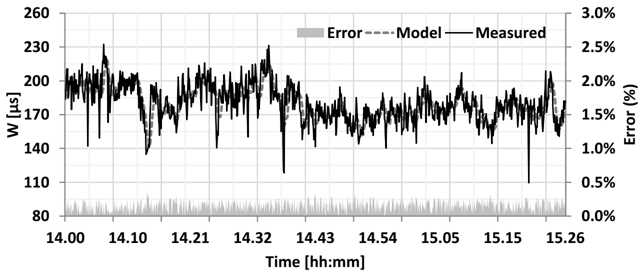

Fig. 14. Average packet waiting time $\hat{W}$ estimated by the model and measured with the test bench for the bidirectional case.

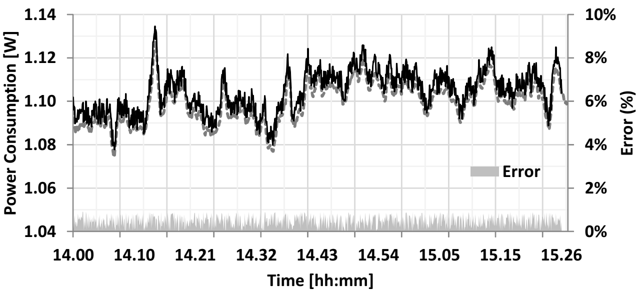

Fig. 13. Average power consumption estimated by the model and measured with the test bench for the bidirectional case.

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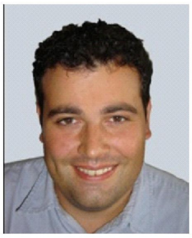

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